

Utilize Network Security Tools

Traffic analysis, vulnerability scanning, and penetration-testing evidence for the DeRexi environment

ASSIGNMENT OBJECTIVE

Document network security tool usage, including at least one Wireshark capture with analysis, one network vulnerability-scanner report, and one network penetration-testing tool output.

Rubric mapping

Required evidence	DeRexi tool / evidence	Status
Wireshark capture + analysis	Wireshark packet capture from an authorized lab or DeRexi test path	Capture still required
Network vulnerability scanner report	Nmap 7.94 vulnerability assessment of fictional DeRexi application server 10.11.2.144	Existing evidence
Network penetration-testing tool output	Social-Engineer Toolkit (SET) controlled credential-harvester lab in Kali Linux	Existing evidence

DeRexi security-testing scope

For this assignment, classroom-lab results are mapped into the fictional DeRexi banking/financial-services environment. The purpose is to show how an analyst would observe traffic, identify exposed services, assess weaknesses, and validate attack paths before deployment.

1. Network vulnerability scanner - Nmap

- Assessment platform: Kali Linux 2023 using Nmap 7.94.
- Discovery scope: 10.11.2.0/24. The host-discovery scan identified 10 active hosts.
- Selected asset: 10.11.2.144, treated in the DeRexi scenario as the internal application server.
- Full TCP service scan: `nmap -sV -sC -p1-65535 10.11.2.144`.
- Observed exposure: TCP/22 open for SSH; 65,534 TCP ports reported closed.
- Vulnerability-oriented scan: `nmap -sC --script=vuln 10.11.2.144`.
- Result: no confirmed target-specific vulnerability was established in the captured output.

DEREXI INTERPRETATION

SSH is a legitimate encrypted administration service, but an exposed remote-management port increases attack surface. For DeRexi, administrative access should be restricted to authorized administrators and trusted management networks, with strong authentication, current patching, and monitoring of authentication activity.

2. Penetration-testing tool output - SET

The Social-Engineer Toolkit (SET) was used in an authorized Kali Linux classroom exercise to demonstrate a controlled credential-harvesting workflow with fake test data.

- Tool: Social-Engineer Toolkit (SET) in Kali Linux.
- Technique: Website Attack Vectors -> Credential Harvester Attack Method.
- Controlled callback / POST address: 10.11.2.97.
- SET launched a local web service on TCP port 80 using a built-in template.
- Observed web activity included HTTP GET requests, a form POST, and an HTTP 302 redirect response.
- Only intentionally fake credentials were used during validation.

DEREXI SCENARIO

In the DeRexi context, the exercise represents how an employee could be deceived by a login-style page that imitates a trusted service. The defensive lesson is to combine user awareness with MFA, password managers, domain/URL verification, browser protections, secure email controls, and monitoring for suspicious credential submissions.

3. Wireshark capture + analysis

This is the only rubric item for which an actual capture is not present in the prior submitted evidence. The final assignment should include one real Wireshark screenshot from an authorized environment and a short packet analysis.

INSERT WIRESHARK CAPTURE HERE

Use a screenshot that clearly shows the packet list and selected packet details.

Capture context	Identify the authorized interface/test and what traffic was intentionally generated.
Source / destination	Record the relevant IP addresses or endpoints visible in the selected packet.
Protocol	Identify the protocol shown (for example TCP, DNS, TLS, HTTP, ICMP).
Ports	Note any relevant source/destination ports and what service they represent.
Packet behavior	Explain what the selected packet is doing - request, response, handshake, lookup, transfer, etc.
Security meaning	Explain what the traffic reveals about DeRexi's security posture or normal network behavior.

SUBMISSION NOTE

Do not fabricate the Wireshark screenshot. Once the capture is inserted and analyzed, the report will address all three rubric requirements. The Nmap and SET sections reuse already documented authorized classroom-lab evidence.